M3503 WEEK 7 2022

SOLVING INITIAL VALUE PROBLEMS WITH LAPLACE TRASFORM.

STEPS :- Take Laplace transform on both sides of the equation.

- Use properties of Laplace transform and initial conditions to get the equation for the Laplace transform of the solution, and solve the equation for the Laplace transform.

- Determine the inverse Laplace transform.

Examples

Solve

① $y'' - 2y' + 5y = 0$; $y(0) = 2$, $y'(0) = 4$

② $y'' + y = t^2 + 2$; $y(0) = 1$, $y'(0) = 7$

③ $y''' - y'' + y' - y = 0$; $y(0) = 1$, $y'(0) = 1$, $y''(0) = 3$

④ $y'' + 3ty' - 6y = 1$; $y(0) = 0$, $y'(0) = 0$

⑤ $y'' - y' - 2y = -8\cos(t) - 2\sin(t)$; $y(\pi/2) = 1$, $y'(\pi/2) = 0$

Solu

① $y'' - 2y' + 5y = 0$

$\Rightarrow \mathcal{L}\{y''\} - 2\mathcal{L}\{y'\} + 5\mathcal{L}\{y\} = \mathcal{L}\{0\}$

Let $\mathcal{L}\{y\} = Y(s)$, then

$\mathcal{L}\{y''\} = s^2 Y(s) - sy(0) - y'(0) = s^2 Y(s) - 2s - 4$

$\mathcal{L}\{y'\} = sY(s) - y(0) = sY(s) - 2$

Substituting into the equation we get

$$s^2Y(s) - 2s - 4 - 2sY(s) + 4 + 5Y(s) = 0$$

$$\Rightarrow Y(s)(s^2 - 2s + 5) = 2s$$

$$\Rightarrow Y(s) = \frac{2s}{s^2 - 2s + 5}$$

$$= \frac{2(s-1)}{(s-1)^2 + 2^2} + \frac{2}{(s-1)^2 + 2^2}$$

$$\Rightarrow y = \mathcal{L}^{-1}\left\{ \frac{2(s-1)}{(s-1)^2+2^2} + \frac{2}{(s-1)^2+2^2} \right\}$$

$$= 2e^t \cos(2t) + e^t \sin(2t)$$

(2). $y'' + y = t^2 + 2$

$$\Rightarrow \mathcal{L}\{y''\} + \mathcal{L}\{y\} = \mathcal{L}\{t^2+2\}$$

$$\Rightarrow s^2Y(s) - sy(0) - y'(0) + Y(s) = \frac{2}{s^3} + \frac{2}{s}$$

$$\Rightarrow Y(s)(s^2+1) - s - 7 = \frac{2}{s^3} + \frac{2}{s}$$

$$\Rightarrow Y(s) = \frac{2 + 2s^2 + s^4 + 7s^3}{s^3(s^2+1)}$$

$$= \frac{2}{s^3} + \frac{s+7}{s^2+1}$$

$$\Rightarrow y = \mathcal{L}^{-1}\left\{ \frac{2}{s^3} + \frac{s+7}{s^2+1} \right\}$$

$$= \mathcal{L}^{-1}\left\{ \frac{2}{s^3} + \frac{s}{s^2+1} + 7\,\frac{1}{s^2+1} \right\}$$

$$= t^2 + \cos(t) + 7\sin(t)$$

③. $y''' - y'' + y' - y = 0$

$\Rightarrow \mathcal{L}\{y'''\} - \mathcal{L}\{y''\} + \mathcal{L}\{y'\} - \mathcal{L}\{y\} = \mathcal{L}\{0\}$

$\Rightarrow s^3Y(s) - s^2y(0) - sy'(0) - y''(0) - s^2Y(s) + sy(0) + y'(0) + sY(s) - y(0) - Y(s) = 0$

$\Rightarrow Y(s)[s^3 - s^2 + s - 1] - s^2 - s - 3 + s = 0$

$\Rightarrow Y(s) = \dfrac{s^2+3}{s^3 - s^2 + s - 1}$

$= \dfrac{s^2+3}{(s^2+1)(s-1)}$

$= \dfrac{-s-1}{s^2+1} + \dfrac{2}{s-1}$

$\Rightarrow y = \mathcal{L}^{-1}\left\{ \dfrac{-s}{s^2+1} - \dfrac{1}{s^2+1} + 2\dfrac{1}{s-1} \right\}$

$= -\cos(t) - \sin(t) + 2e^t$

⑭. $y'' + 3ty' - 6y = 1$

$\Rightarrow \mathcal{L}\{y''\} + 3\mathcal{L}\{ty'\} - 6\mathcal{L}\{y\} = \mathcal{L}\{1\}$

$\Rightarrow s^2Y(s) - sy(0) - y'(0) + 3\left[-\dfrac{d}{ds}(sY(s) - y(0))\right] - 6Y(s) = \dfrac{1}{s}$

$\Rightarrow s^2Y(s) + 3[-Y(s) - sY'(s)] - 6Y(s) = \dfrac{1}{s}$

$\Rightarrow Y'(s) - \dfrac{Y(s)(s^2-9)}{3s} = -\dfrac{1}{3s^2}$

Integrating factor $= \exp\left(\int \dfrac{-(s^2-9)}{3s}\, ds\right)$

$= \exp\left(-\int \left(\dfrac{s}{3} - \dfrac{3}{s}\right) ds\right)$

$= \exp\left(-s^2/6 + 3\ln s\right)$

$= s^3 e^{-s^2/6}$

So, $\dfrac{d}{ds}\left(Y(s)\, s^3 e^{-s^2/6}\right) = -\dfrac{s}{3} e^{-s^2/6}$

$$\Rightarrow Y(s)\, s^3 e^{-s^2/6} = e^{-s^2/6} + C$$

$$\Rightarrow Y(s) = \frac{1}{s^3} + \frac{Ce^{s^2/6}}{s^3}$$

Since $\lim_{s\to\infty} Y(s) = 0$, which can only be possible if $C = 0$, we therefore have

$$Y(s) = \frac{1}{s^3}$$

$$\Rightarrow y = \mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\}$$

$$= \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^3}\right\}$$

$$= \frac{t^2}{2}$$

⑤. $y'' - y' - 2y = -8\cos(t) - 2\sin(t)$; $y(\pi/2) = 1$, $y'(\pi/2) = 0$

We cannot apply Laplace transform right away since the given initial conditions are not for $t = 0$. To get initial conditions for $t = 0$, we set

$$W(t) = y(t + \pi/2)$$

$$\Rightarrow W'(t) = y'(t + \pi/2)$$

$$\Rightarrow W''(t) = y''(t + \pi/2)$$

Also, $W(0) = y(0 + \pi/2) = y(\pi/2) = 1$

$$W'(0) = y'(0 + \pi/2) = y'(\pi/2) = 0$$

Replacing t by $(t + \pi/2)$ in the given equation, we get

$$W''(t) - W'(t) - 2W(t) = 8\sin(t) - 2\cos(t),\ W(0) = 1,\ W'(0) = 0$$

$$\Rightarrow \mathcal{L}\{w''(t)\} - \mathcal{L}\{w'(t)\} - 2\mathcal{L}\{w(t)\} = 8\mathcal{L}\{\sin(t)\} - 2\mathcal{L}\{\cos t\}$$

$$\Rightarrow s^2 Y(s) - s w(0) - w'(0) - sY(s) + w(0) - 2Y(s) = \frac{8}{s^2+1} - \frac{2s}{s^2+1}$$

$$\Rightarrow Y(s)(s^2 - s - 2) - s + 1 = \frac{8 - 2s}{s^2+1}$$

$$\Rightarrow Y(s) = \frac{s^3 - s^2 - s + 7}{(s^2+1)(s-2)(s+1)}$$

$$= \frac{7}{5}\left(\frac{s}{s^2+1}\right) - \frac{11}{5}\left(\frac{1}{s^2+1}\right) + \frac{3}{5}\left(\frac{1}{s-2}\right) - \frac{1}{s+1}$$

$$\Rightarrow w(t) = \mathcal{L}^{-1}\left\{\frac{7}{5}\left(\frac{s}{s^2+1}\right) - \frac{11}{5}\left(\frac{1}{s^2+1}\right) + \frac{3}{5}\left(\frac{1}{s-2}\right) - \frac{1}{s+1}\right\}$$

$$= \frac{7}{5}\cos(t) - \frac{11}{5}\sin(t) + \frac{3}{5}e^{2t} - e^{-t}$$

Since $w(t) = y(t + \pi/2)$, $y(t) = w(t - \pi/2)$.

Replacing t by $(t - \pi/2)$ in the solutions gives

$$y(t) = \frac{7}{5}\cos(t - \pi/2) - \frac{11}{5}\sin(t - \pi/2) + \frac{3}{5}e^{2(t-\pi/2)} - e^{-(t-\pi/2)}$$

$$= \frac{7}{5}\sin(t) + \frac{11}{5}\cos(t) + \frac{3}{5}e^{2t-\pi} - e^{\pi/2 - t}$$

LAPLACE TRANSFORM OF PIECEWISE CONTINUOUS FUNCTIONS

The following functions are useful in studying the Laplace transform of piecewise functions

The functions $u(t-a) = \begin{cases} 0, & 0 < t < a \\ 1, & t > a \end{cases}$

is called a unit step function.

The function $\Pi_{a,b}(t) = \begin{cases} 0, & t < a \\ 1, & a < t < b \\ 0, & t > b \end{cases}$

$= u(t-a) - u(t-b)$

is called the rectangular window function

Any piecewise continuous function can be expressed in terms of a unit step function

Examples

Express the following functions in terms of the unit step function

① $f(t) = \begin{cases} 2, & 0 \le t < 3 \\ -2, & t \ge 3 \end{cases}$

② $f(t) = \begin{cases} 1, & 0 < t < 1 \\ -1, & 1 < t < 2 \\ 0, & t > 2 \end{cases}$

③ $f(t) = \begin{cases} 0, & 0 < t < 1 \\ 2, & 1 < t < 2 \\ 1, & 2 < t < 3 \\ 3, & t < 3 \end{cases}$

Sol'n

① $f(t) = 2[1 - u(t-3)] - 2u(t-3)$

$= 2 - 4u(t-3)$

② $f(t) = 1 - u(t-1) - [u(t-1) - u(t-2)]$

$= 1 - 2u(t-1) + u(t-2)$

③ $f(t) = 2[u(t-1) - u(t-2)] + u(t-2) - u(t-3) + 3u(t-3)$

$= 2u(t-1) - 2u(t-2) + u(t-2) + 2u(t-3)$

$= 2u(t-1) - u(t-2) + 2u(t-3)$

Using the definition of Laplace transform, it can be shown that

$$\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}$$

SECOND SHIFT THEOREM

If $\mathcal{L}\{f(t)\} = F(s)$ exists for $s > \alpha \geq 0$ and $a > 0$, then

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$$

Examples

1. Find the Laplace transform of

(a) $f(t) = 2u(t-1) - u(t-2) + 2u(t-3)$

(b) $f(t) = \begin{cases} 0, & 0 < t < 1 \\ t, & 1 < t < 2 \\ 1, & t > 2 \end{cases}$

2. Determine the Inverse Laplace transform of

(a) $F(s) = \dfrac{e^{-2s}}{s-1}$

(b) $F(s) = \dfrac{e^{-3s}(s-5^{-})}{(s+1)(s+2)}$

3. The current $I(t)$ in an RLC series circuit is governed by the initial value problem

$$I''(t) + 2I'(t) + 2I(t) = g(t);$$

$$I(0) = 10,\ I'(0) = 0,$$

where $g(t) = \begin{cases} 20, & 0 < t < 3\pi \\ 0, & 3\pi < t < 4\pi \\ 20, & 4\pi < t \end{cases}$

Determine the current as a function of t

Solu

1. (a) $\mathcal{L}\{2u(t-1) - u(t-2) + 2u(t-3)\}$

$= 2\mathcal{L}\{u(t-1)\} - \mathcal{L}\{u(t-2)\} + 2\mathcal{L}\{u(t-3)\}$

$= 2\dfrac{e^{-s}}{s} - \dfrac{e^{-2s}}{s} + \dfrac{2e^{-3s}}{s}$

(b) $\mathcal{L}\{f(t)\} = \mathcal{L}\{tu(t-1) - tu(t-2) + u(t-2)\}$

$= \mathcal{L}\{tu(t-1)\} + \mathcal{L}\{(1-t)u(t-2)\}$

$= e^{-s}\mathcal{L}\{t+1\} + e^{-2s}\mathcal{L}\{-t-1\}$

$= e^{-s}\left[\dfrac{1}{s^2} + \dfrac{1}{s}\right] + e^{-2s}\left[-\dfrac{1}{s^2} - \dfrac{1}{s}\right]$

2. (a) $\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s-1}\right\}$

$= u(t-2)e^{t-2}$ since $\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} = e^t$

$= \begin{cases} 0, & 0 < t < 2 \\ e^{t-2}, & t > 2 \end{cases}$

(b) $\mathcal{L}^{-1}\left\{\frac{s-5}{(s+1)(s+2)}\right\} = 7\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} - 6\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}$

$= 7e^{-2t} - 6e^{-t}$

Thus, $\mathcal{L}^{-1}\left\{\frac{e^{-3s}(s-5)}{(s+1)(s+2)}\right\} = u(t-3)\left[7e^{-2(t-3)} - 6e^{-(t-3)}\right]$

$= \begin{cases} 0, & 0 < t < 3 \\ 7e^{-2(t-3)} - 6e^{-(t-3)}, & t > 3 \end{cases}$

3. $I'' + 2I' + 2I = 20 - 20u(t-3\pi) + 20u(t-4\pi)$

$\Rightarrow s^2Y(s) - 10s + 2sY(s) - 20 + 2Y(s) = \frac{20}{s} - 20\frac{e^{-3\pi s}}{s} + 20\frac{e^{-4\pi s}}{s}$

$\Rightarrow Y(s)\left[s^2+2s+2\right] = 20\left[\frac{1 - e^{-3\pi s} + e^{-4\pi s}}{s}\right] + 10s + 20$

$\Rightarrow Y(s) = 10(e^{-4\pi s} - e^{-3\pi s})\left[\frac{1}{s} - \frac{s+1}{(s+1)^2+1} - \frac{1}{(s+1)^2+1}\right] + \frac{10}{s}$

$\Rightarrow I(t) = 10u(t-4\pi)\left[1 - e^{-(t-4\pi)}(\cos(t-4\pi) + \sin(t-4\pi))\right]$

$- 10u(t-3\pi)\left[1 - e^{-(t-3\pi)}(\cos(t-3\pi) + \sin(t-3\pi))\right]$

$+ 10$

$= 10u(t-4\pi)\left[1 - e^{-(t-4\pi)}(\cos t + \sin t)\right]$

$- 10u(t-3\pi)\left[1 + e^{-(t-3\pi)}(\cos t + \sin t)\right]$

$+ 10$